Type MS · Define diffusion, osmosis and active transport

Answer 1 of 9 · Question 2 part b · 3 marks

2021 NOV · Paper 01 · find part b in this mark scheme extract

2 (a)	• movement of molecules; • from high to low concentration/down a diffusion/concentration gradient;		2marks
(b)	any three of • movement of molecules against a concentration gradient/low to high concentration; • using energy; • in the form of ATP; • ref protein channels;		3marks
(c)	any two of • movement of water; • from high to low water potential/down a water potential gradient; • across a partially/selectively/semi permeable membrane;	ignore ref to high/low conc of water	2marks

Total 7 marks

Question number	Answer	Notes	Marks

Type MS · Define diffusion, osmosis and active transport

Answer 2 of 9 · Question 5 part a · 2 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

5(a)	movement of a substance against a concentration gradient/low to high concentration; using energy/ATP;		**(2)**

Question Number	Answer	Additional guidance	Mark
5(b)	• cut pieces of potato to same size/length/mass; • weigh each piece; • place into different solutions; • leave for at least 30 minutes; • remove and dry; • re-weigh/re-measure length; • determine which concentration has no / least change in mass/length;		**(5)**

Question Number	Answer	Additional guidance	Mark
5(c)(i)	0.10 × 100; 7.90 1.3;	Ecf allow correct answer to incorrect figures used in calculation Allow 1.26 / 1.27 for final correct answer	**(2)**

Type MS · Define diffusion, osmosis and active transport

Answer 2 of 9 · Question 5 part a · 2 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Question Number	Answer	Additional guidance	Mark
5(c)(ii)	initial mass/volume/size varies; greater change in mass/volume if potato is bigger; provides a standardised result; allows comparison;		**(2)**
Question Number	Answer	Additional guidance	Mark
5(c)(iii)	• plot a graph; • concentration vs percentage change; • find (from graph) concentration at which percentage change is zero/lowest change; • this is the concentration of the contents;		**(4)**

Total 15 Marks

Type MS · Define diffusion, osmosis and active transport

Answer 2 of 9 · Question 5 part a · 2 marks

2019 JUN · Paper 02 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

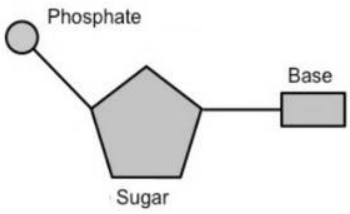
Paper 2: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Define diffusion, osmosis and active transport

Answer 3 of 9 · Question 8 part a.i · 2 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

8 (a) (i)	Any two from: • (movement of substances/named solute or gas) against a concentration gradient(1) • using energy/ATP(1)		2
(ii)	Any two from: • parent homozygous recessive/heterozygous / carriers (1) • one (recessive) allele inherited from each parent (1) • reference to homozygous recessive disorder/caused by recessive alleles (1)		2
(iii)	• protein/CFTR is not modified/folded/protein has an incorrect shape(1) • protein/CFTR not transported (to its destination)(1)		2
(iv)	1408 x 3 (1) 4224	Full marks for correct final answer. Max 2 marks ecf for final answer if it is correct from calculation	2
(v)	• (deoxyribose) sugar(1) • joined to a (nitrogenous) base/named base(1) • phosphate group attached to sugar (1) 	Allow one mark only if all 3 components just listed Fully labelled diagram for full marks	3
(b)	any three from • deletion(1)	allow description of deletion	3

Type MS · Define diffusion, osmosis and active transport

Answer 3 of 9 · Question 8 part a.i · 2 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

	• of one codon/TTT (1) • phenylalanine missing from protein (1) • shape of final protein changed/different protein formed (1)		

Total for Question 8 = 14 Marks

Type MS · Define diffusion, osmosis and active transport

Answer 4 of 9 · Question 2 part a · 2 marks

2021 NOV · Paper 01 · find part a in this mark scheme extract

2 (a)	• movement of molecules; • from high to low concentration/down a diffusion/concentration gradient;		2marks
(b)	any three of • movement of molecules against a concentration gradient/low to high concentration; • using energy; • in the form of ATP; • ref protein channels;		3marks
(c)	any two of • movement of water; • from high to low water potential/down a water potential gradient; • across a partially/selectively/semi permeable membrane;	ignore ref to high/low conc of water	2marks

Total 7 marks

Question number	Answer	Notes	Marks

Type MS · Define diffusion, osmosis and active transport

Answer 5 of 9 · Question 2 part c · 2 marks

2021 NOV · Paper 01 · find part c in this mark scheme extract

2 (a)	• movement of molecules; • from high to low concentration/down a diffusion/concentration gradient;		2marks
(b)	any three of • movement of molecules against a concentration gradient/low to high concentration; • using energy; • in the form of ATP; • ref protein channels;		3marks
(c)	any two of • movement of water; • from high to low water potential/down a water potential gradient; • across a partially/selectively/semi permeable membrane;	ignore ref to high/low conc of water	2marks

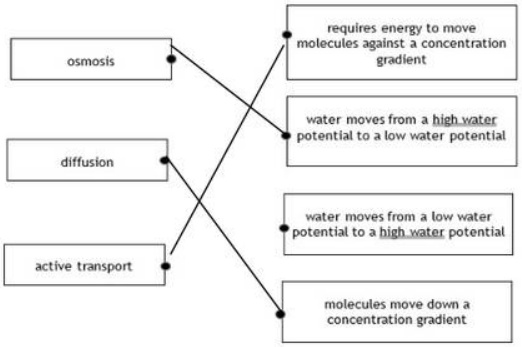
Total 7 marks

Question number	Answer	Notes	Marks

Type MS · Define diffusion, osmosis and active transport

Answer 6 of 9 · Question 1 part b · 3 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

Number				
1	(a) (i)	X = cell membrane; Y = nucleus; Z = cytoplasm;	Allow nuclear membrane	3
	(ii)	mitochondria/ribosomes/endoplasmic reticulum		1
	(iii)	• similar cells work together to form tissues; • (different) tissues work together to form organs;		2
	(b)		Reject more than one line from each process	3
	(c) (i)	A = squamous (epithelium) B = ciliated (epithelium)		2
	(ii)	60 x 40; 2400 (µm) (÷1000); 2.4 (mm);	Ecf from mp1. Max 2 marks e.g. 60 ÷ 40 (÷1000) = 0.0015	3

Total 14 marks

Type MS · Define diffusion, osmosis and active transport

Answer 6 of 9 · Question 1 part b · 3 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Define diffusion, osmosis and active transport

Answer 7 of 9 · Question 3 part a.iii · 1 mark

2023 JAN · Paper 02 · extract 1 of 2 · find part a.iii in this mark scheme extract

Question number					

3	(a)	(i)	surface area = 24 (mm^2);
			volume = 8 (mm^3);
			3:1;
				Full marks for correct final answer	3

		(ii)	C 92;		1
		(iii)	movement from a high concentration to a low concentration;		1
		(iv)	• reduces the time taken to become colourless/rate of diffusion/reaction is faster; • the diffusion distance is shorter/greater area over which diffusion/collisions can occur;		2
	(b)	(i)	• use acid heated to different temperatures/idea of different temperatures; • compare/measure the time taken to become colourless/compare rate of diffusion;		2
		(ii)	Any two from: • the side length of the cube/size of the cube/surface area/volume of cube; • acid concentration; • concentration of alkali/indicator in the cube;	accept volume of acid Allow volume of indicator	2

Total 11 marks

Type MS · Define diffusion, osmosis and active transport

Answer 7 of 9 · Question 3 part a.iii · 1 mark

2023 JAN · Paper 02 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Define diffusion, osmosis and active transport

Answer 8 of 9 · Question 1 part whole · 6 marks

2023 JUN · Paper 01 · find part whole in this mark scheme extract

1	water; selectively; down; volume; ATP; against;		6

Total 6 marks

Question number	Answer	Notes	Marks

Type MS · Define diffusion, osmosis and active transport

Answer 9 of 9 · Question 7 part b · 2 marks

2024 NOV · Paper 01 · find part b in this mark scheme extract

7 (a)	• measure initial mass (of food)use same mass of food; • measure initial temperature of water; • burn food; • take final temperature of water; • repeat for each food; • compare temperature difference (of water)/results for each food;		6
(b)	• active transport/uptake; • reference to low to high concentration/against concentration gradient;		2

Total Marks 8

Question number	Answer	Notes	Marks